CLASS 11-CBSE MATHEMATICS

CHAPTER: INTRODUCTION TO 3D GEOMETRY-SET-1

TIME : 1Hr **Mx.Marks:30**

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. Name the octant in which (i)(-2,6,7) and (ii)(9,-1,-2) lies.

2. Find image of (–2, 3, 5) in YZ plane.

3. Find the distance of point P(3, –2,1) from z–axis.

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. Find the distance between the points P and Q having coordinates (–2, 3, 1) and (2, 1, 2).

5. Find the ratio in which XOZ-plane divides the join of (2, 3, 1) and (6, 7, 1)

6. Find the point on y–axis which is equidistant from the point (3, 1, 2) and (5, 5, 2).

7. Find the coordinates of the point P which is five-sixth of the way from A(2, 3, –4) to B(8, –3, 2).

8. Find the centroid of a triangle, mid-points of whose sides are (1, 2, –3), (3, 0, 1) and (–1, 1, –4).

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. Show that the points (0, 7, 10), (−1, 6, 6) and (−4, 9, 6) are the vertices of an isosceles right-angled triangle.

10. Show that the points $A(1, 3, 4)$, $B(-1, 6, 10)$, $C(-7, 4, 7)$ and $D(-5, 1, 1)$ are the vertices of a rhombus.

11. Find the locus of the point, the sum of whose distances from the points $A(4, 0, 0)$ and $B(-4, 0, 0)$ is equal to 10.

SECTION D

Q.IV Answer the following questions **[4X2 = 8 MARKS]**

12. If A and B be the points (3, 4, 5) and (−1, 3, −7), respectively, find the equation of the set of points P such that $PA^2 + PB^2 = k^2$, where k is a constant.

13. Planes are drawn parallel to the coordinate planes through the points (3, 0, −1) and (−2, 5, 4). Find the lengths of the edges of the parallelepiped so formed.